

SABIN PRADHAN

AI ENGINEER

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SUMMARY

AI Systems Engineer specializing in scalable LLM architecture, advanced RAG pipelines, and production-ready AI infrastructure. Proven track record of architecting end-to-end GenAI systems optimized for latency, token efficiency, and deterministic outputs. Passionate about bridging the gap between AI prototypes and high-performance, enterprise-grade deployment.

SKILLS

- **AI Ecosystem & LLMs:** LangChain, OpenAI API, Anthropic Claude, OpenRouter, Hugging Face (Transformers, PEFT, TRL), Sentence Transformers, MediaPipe.
- **GenAI Architecture:** RAG Pipelines, Agentic Workflows, Tool Calling, Semantic Search, Embedding Models, JSON Structured Outputs, LLM Fine-Tuning (QLoRA, 4-bit Quantization).
- **Backend & Cloud:** Python (FastAPI), TypeScript (Node.js), AWS / Cloud Deployments, REST APIs, Stateless Architecture.
- **Data & Vector DBs:** PostgreSQL (pgvector), FAISS, Qdrant, MongoDB, Redis (Caching).
- **MLOps & Infra:** Docker, CI/CD (GitHub Actions), LLM Observability, Cloud GPU Training, Git, Next.js / React.

PROJECTS

GenAI Customer Support Agent (RAG + Tool Calling)

FastAPI, LangChain, OpenRouter

- Architected a production-grade AI support agent capable of resolving multi-turn e-commerce queries by integrating dynamic RAG and external tool-calling.
- Engineered an end-to-end document ingestion pipeline, achieving high-precision semantic retrieval through optimized chunking strategies and vector embeddings.
- Implemented a robust API-driven execution layer, automating order lookups while enforcing strict JSON schema validation for deterministic backend processing.
- Reduced AI hallucination rates by ~30% by rigorously grounding LLM responses in real-time product databases and proprietary knowledge graphs.

Gym Coach AI (Personalized RAG System)

Python, FAISS, PyMuPDF

- Engineered a personalized AI advisory system, leveraging FAISS and PyMuPDF to extract and retrieve dense context from vectorized fitness literature.
- Architected a multi-step semantic retrieval pipeline capable of generating highly targeted, knowledge-grounded responses dynamically adapted to user profiles.
- Optimized inference latency by 40% and minimized token costs by implementing semantic caching and routing queries through task-specific LLMs.

Text-to-Sign AI Accessibility Tool

Python, MediaPipe, Sentence Transformers

- Engineered a text-to-sign pipeline using semantic search and embeddings to retrieve Auslan signs and generate 2D avatar animations, with fingerspelling fallback for unsupported words.

LLM Fine-Tuning: Qwen3-14B with QLoRA

Hugging Face Transformers, PEFT, TRL

- Fine-tuned a 14-billion-parameter language model on UltraChat data using 4-bit QLoRA on a single rented NVIDIA RTX 5090 cloud GPU, reducing trainable parameters and memory enough to adapt a model of this size on one GPU.

EDUCATION

Master of Information Technology (Artificial Intelligence)

2026 – 2027

Charles Darwin University, Darwin, NT

Bachelor of Information Technology

2023 – 2025

Victoria University, Sydney, NSW